

Practical: Flight Information System using HBase and HiveQL

This practical demonstrates how to create and manage a **Flight Information System** using **HBase** and **HiveQL**.

Aim

To implement a Flight Information System using HBase and Hive that includes:

1. Creating Flight Information table in HBase
 2. Demonstrating Create, Alter, Drop operations in HBase
 3. Connecting HBase table with Hive External Table
 4. Finding Total Departure Delay using HiveQL
 5. Finding Average Departure Delay using HiveQL
 6. Creating Index on Flight Information Table
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Software Requirements

- Hadoop
 - HBase
 - Hive
 - Java
 - Linux/Ubuntu Terminal
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Step 1: Start Hadoop, HBase, and Hive Services

Start Hadoop

```
start-dfs.sh  
start-yarn.sh
```

Start HBase

```
start-hbase.sh
```

Open HBase Shell

hbase shell

Step 2: Create Flight Information Table in HBase

We will create a table named flight_info.

Create Table

```
create 'flight_info',  
'flight_details',  
'schedule',  
'delay'
```

Explanation

- flight_details → Flight ID, Airline, Source, Destination
 - schedule → Departure and Arrival timings
 - delay → Delay information
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Step 3: Insert Data into HBase Table

Insert Flight Records

```
put 'flight_info','F101','flight_details:airline','AirIndia'  
put 'flight_info','F101','flight_details:source','Mumbai'  
put 'flight_info','F101','flight_details:destination','Delhi'  
put 'flight_info','F101','schedule:departure','10:00'  
put 'flight_info','F101','schedule:arrival','12:00'  
put 'flight_info','F101','delay:departure_delay','20'  
  
put 'flight_info','F102','flight_details:airline','Indigo'  
put 'flight_info','F102','flight_details:source','Pune'  
put 'flight_info','F102','flight_details:destination','Bangalore'  
put 'flight_info','F102','schedule:departure','14:00'  
put 'flight_info','F102','schedule:arrival','16:00'  
put 'flight_info','F102','delay:departure_delay','10'  
  
put 'flight_info','F103','flight_details:airline','SpiceJet'  
put 'flight_info','F103','flight_details:source','Chennai'
```

```
put 'flight_info','F103','flight_details:destination','Hyderabad'
put 'flight_info','F103','schedule:departure','18:00'
put 'flight_info','F103','schedule:arrival','19:30'
put 'flight_info','F103','delay:departure_delay','15'
```

Step 4: View Table Data

Scan Complete Table

```
scan 'flight_info'
```

Get Specific Record

```
get 'flight_info','F101'
```

Step 5: Demonstrate Create, Alter, and Drop Operations

A. Create Table

```
create 'flight_backup','details'
```

Verify

```
list
```

B. Alter Table

Add new column family status.

```
alter 'flight_info','status'
```

Insert Data into New Column Family

```
put 'flight_info','F101','status:flight_status','Delayed'
```

Verify

```
scan 'flight_info'
```

C. Drop Table

Disable Table First

disable 'flight_backup'

Drop Table

drop 'flight_backup'

Verify

list

Step 6: Open Hive Shell

Exit HBase shell:

exit

Start Hive:

hive

Step 7: Create Hive Database

CREATE DATABASE flightdb;

Use Database

USE flightdb;

Step 8: Create External Hive Table Connected to HBase

```
CREATE EXTERNAL TABLE flight_hive(  
  flight_id STRING,  
  airline STRING,  
  source STRING,  
  destination STRING,  
  departure STRING,  
  arrival STRING,  
  departure_delay INT  
)  
STORED BY 'org.apache.hadoop.hive.hbase.HBaseStorageHandler'
```

```
WITH SERDEPROPERTIES (  
  "hbase.columns.mapping" =  
  ":key,  
  flight_details:airline,  
  flight_details:source,  
  flight_details:destination,  
  schedule:departure,  
  schedule:arrival,  
  delay:departure_delay"  
)  
TBLPROPERTIES ("hbase.table.name" = "flight_info");
```

Step 9: Display Hive Table Data

```
SELECT * FROM flight_hive;
```

Expected Output

flight_id	airline	source	destination	departure	arrival	departure_delay
F101	AirIndia	Mumbai	Delhi	10:00	12:00	20
F102	Indigo	Pune	Bangalore	14:00	16:00	10
F103	SpiceJet	Chennai	Hyderabad	18:00	19:30	15

Step 10: Find Total Departure Delay in Hive

```
SELECT SUM(departure_delay) AS total_delay  
FROM flight_hive;
```

Expected Output

total_delay
45

Step 11: Find Average Departure Delay in Hive

```
SELECT AVG(departure_delay) AS average_delay  
FROM flight_hive;
```

Expected Output

average_delay

15

Step 12: Create Index on Flight Information Table

Create Index

```
CREATE INDEX idx_airline  
ON TABLE flight_hive (airline)  
AS 'COMPACT'  
WITH DEFERRED REBUILD;
```

Rebuild Index

```
ALTER INDEX idx_airline  
ON flight_hive  
REBUILD;
```

Show Indexes

```
SHOW INDEXES ON flight_hive;
```

Step 13: Verification Queries

Find Flights by Airline

```
SELECT * FROM flight_hive  
WHERE airline='Indigo';
```

Find Flights with Delay Greater than 10 Minutes

```
SELECT flight_id, airline, departure_delay  
FROM flight_hive  
WHERE departure_delay > 10;
```

Expected Viva Questions

1. What is HBase?

HBase is a NoSQL distributed database built on Hadoop HDFS for storing large sparse datasets.

2. What is Hive?

Hive is a data warehouse framework that provides SQL-like querying on Hadoop data.

3. What is an External Table in Hive?

An external table maps external data without moving it into Hive warehouse.

4. Why use HBase with Hive?

- HBase provides fast real-time access
 - Hive provides SQL querying capability
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Conclusion

Successfully implemented a Flight Information System using HBase and HiveQL by:

- Creating and managing HBase tables
- Connecting HBase with Hive
- Performing analytical queries
- Creating indexes for faster access